

FINAL EXAMINATION **ANSWERS**

ECON 201, Spring 2014

Prof. Ryan Edwards

May 19, 2014

11am–1pm

Kiely 170

RULES:

- Don't start until I say so. **DON'T LOOK THROUGH THE EXAM.**
- But please check that you have all 10 pages.
- Write your name and **seat number** at the top.
- When I call time, **stop**. If you don't, you get an F.
- No books, no notes, no calculators.
- There are 3 questions worth a total of 90 possible points, one for each minute of time you have. Budget your time accordingly.
- If I catch you cheating in any way, you will receive an F for the course and a referral to the VP for Student Affairs.

Points:	1	2	3	4	5	Total:
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1. (20 points total) *Short answer questions.*

- (a) (5 points) Recall that $MV = PY$. What does this relationship tell you about the long-run rate of inflation if the long-run rate of money growth is 5%, the long-run rate of income growth is 3%, and velocity is not changing? Be specific and show your work.

Answer: Taking growth rates of both sides reveals that

$$g[M] + g[V] = g[P] + g[Y]$$

If we know that $g[M] = 0.05$ while $g[Y] = 0.03$, and $g[V] = 0$, then it must be the case that $g[P] \equiv \pi = g[M] - g[Y] = 0.05 - 0.03 = 0.02$. So the answer is 2%.

- (b) (5 points) Recall the Fisher equation, relating the nominal interest rate i to the real interest rate R and inflation: $i = R + \pi$. What is the lowest real interest rate R the Fed can obtain if $\pi = -0.01$ and why? Do you think this might be beneficial or problematic? Discuss.

Answer: You can't charge a negative nominal interest rate; the "zero lower bound" is $i = 0$, past which we can't go any further lower. So if the Fed can't set i lower than zero, and inflation is -1% , then the real interest rate equals $R = i - \pi = 0 - (-0.01) = 0.01$ or 1% in this world, and it can't go any lower. This might be a problem because when the economy is slumping, the Fed needs to reduce R in order to stimulate investment spending, and if the slump is big enough, $R = 0.01$ might not be low enough to stimulate spending sufficiently. Unless inflation were higher, the economy might not get going again.

Remember that national income accounting suggests that $S - I = NX$, where S is domestic saving, I is domestic investment, and NX is the trade balance. Further, $S = [Y - T - C] + [T - G]$ where T is taxes. Finally, assume NX is constant.

- (c) (5 points) If the government cuts taxes, T , would that help or hurt investment, I , or would it do nothing at all? Explain.

Answer: If the government cuts taxes, that raises private spending, $Y - T - C$, but it reduces public spending, $T - G$, by an equal amount. The net effect on national saving is zero (unless the tax cut changes consumption C or income Y). So in the simplest view, a tax cut would do nothing at all to investment.

- (d) (5 points) If the government cuts spending, G , would that help or hurt investment, I , or would it do nothing at all? Explain.

Answer: The only place G shows up is in public saving, $T - G$, which would unambiguously rise if G were to fall. Absent another reaction, a cut in G must produce a rise in investment because national saving must have risen.

2. (34 points total) *How Solow can you go?*

Imagine the U.S. is a Solow economy in which the saving rate is $\bar{s}$, depreciation is $\bar{d}$, capital is K , population is L , total factor productivity is $\bar{A}$, and GDP is

$$Y = F(\bar{A}, K, L) = \bar{A}K^{1/3}L^{2/3}. \quad (1)$$

Capital is accumulated according to the standard flow equation, where the change in the capital stock is equal to saving minus depreciation,

$$\Delta K = \bar{s}Y - \bar{d}K. \quad (2)$$

Finally, the wage equals the marginal product of labor, given by

$$w = MPL = \frac{2}{3}\bar{A}\left(\frac{K}{L}\right)^{1/3} \quad (3)$$

- (a) (2 points) Recall the definition of the steady state. What does it imply about equation (2)? Write it down and describe.

Answer: In the steady state, the change in the capital stock must be zero; equation (2) equals zero.

$$\Delta K = sY - dK = 0.$$

- (b) (5 points) Using your answer to part (a), substitute for Y using equation (1), and solve for K^* , the level of capital in the steady state, as a function of s , d , A , and L .

Answer:

$$\begin{aligned} sY - dK &= 0 \\ sAK^{1/3}L^{2/3} - dK &= 0 \\ sAK^{1/3}L^{2/3} &= dK \\ sAL^{2/3} &= dK^{2/3} \\ \frac{sA}{d}L^{2/3} &= K^{2/3} \\ K^* &= \left(\frac{sA}{d}\right)^{3/2} L \end{aligned}$$

- (c) (3 points) Take the growth rate of both sides of your answer to part (b) and simplify.

Answer:

$$g[K^*] = g\left[\left(\frac{sA}{d}\right)^{3/2}\right] + g[L] = \frac{3}{2}(g[s] + g[A] - g[d]) + g[L]$$

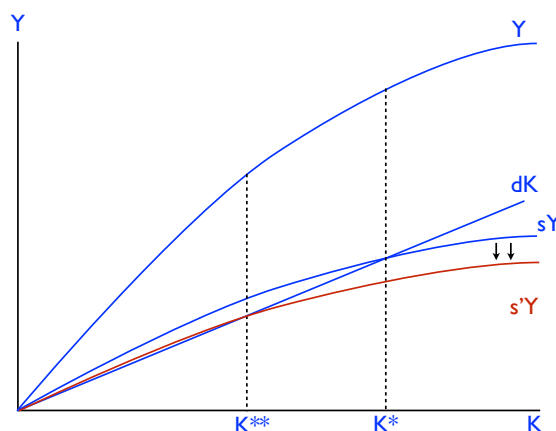
What determines the saving rate, s ? A bunch of things, probably; family size and composition matter, and so does education and probably wealth. What about *taxes on capital*, which raise the “price” of saving? (When capital taxes are higher, the price of saving is higher.)

If we were in Econ 202 right now, we’d have to worry about income versus substitution effects. But in Econ 201, we typically assume that substitution effects dominate, and we’ll do so here too. That means that when the price of something rises, we’ll assume that people substitute away from it.

- (d) (3 points) Using this logic, would the saving rate rise, fall, or stay the same if the tax rate on capital was higher? Explain fully.

If the substitution effect dominates, then raising the price of saving by raising the tax on capital will reduce the saving rate.

- (e) (4 points) Illustrate what you have described in part (d) using the Solow diagram below. Show the effects, if any, of an increase in the tax on capital. Explain fully.



If a tax on capital reduces the saving rate, then we would see a downward shift in the saving curve, produces a lower steady-state level of capital at K^{**} .

We could also tax labor instead of capital. A tax on labor reduces the price of leisure and effectively raises the price of labor.

- (f) (3 points) Using this logic, would labor supplied rise, fall, or stay the same if the tax rate on labor was higher? Explain fully.

Similarly, we would expect labor supply to fall if the tax rate on labor were to be set higher. As long as substitution effects dominate, then a reduction in the price of leisure, which is the wage, produces an increase in the demand for leisure, or a reduction in labor supply.

If we can raise revenue either by taxing labor or capital, how do we decide which one is better to tax? One perspective is to compare reductions in the steady-state capital stock K^* that result.

- (g) (4 points) Using your answer to part (c), identify which reduces steady-state K^* more: a 1% reduction in the saving rate, s , or a 1% reduction in labor supplied, L . Explain.

This is a little unfair, because we don't know how much the saving rate would change in response to a capital tax. But we also don't know how much labor will respond either. As part (c) reveals, a 1% reduction in the saving rate will produce a 1.5% reduction in steady-state capital. By contrast, a 1% reduction in labor supplied only produces a 1% reduction in steady-state capital. Again, this doesn't tell us a lot without a better sense of how s and L respond to taxes, but at least it's something.

- (h) (5 points) Suppose we knew that labor supply was much less responsive to taxes, or *more inelastic*, than saving. In which way would this tilt the decision, toward taxing labor or taxing capital, or would it leave the decision unaffected? Explain.

If labor supply doesn't respond to taxes, we're saying that taxing labor supply won't reduce it very much. This would tend to tilt the decision toward taxing labor, since doing so wouldn't produce as much of a change in behavior, here labor supply. The cost of taxing labor would be lower in such a scenario than if labor responded strongly to taxes.

- (i) (5 points) Suppose we moved from theory to real-world analysis, and in so doing we recognized that many workers do not own capital at all, and thus are relatively poor, but they do supply labor. Meanwhile, those who own capital tend to be considerably more wealthy than those who do not. In which way would this tilt the decision, toward taxing labor or taxing capital, or would it leave the decision unaffected? Explain.

Here I'm trying to get at the notion of equity or equality. If most people supply labor but don't own capital, and if those who own capital tend to be the wealthiest, this fact would probably push us toward taxing capital and away from taxing labor. This is clearly a debatable point, but most folks would probably argue that the rich can better afford to be taxed.

3. (36 points total) *Thanks a lot, Generation X, You Broke the Short Run.*

“Generation X” is the term for Americans born roughly between the 1960s and 1980s. People born later are commonly called “Millennials.” Recall the standard IS, MP, and PC Curves we saw in class:

$$\tilde{Y} = \bar{a} - \bar{b}(R - \bar{r}) \quad (4)$$

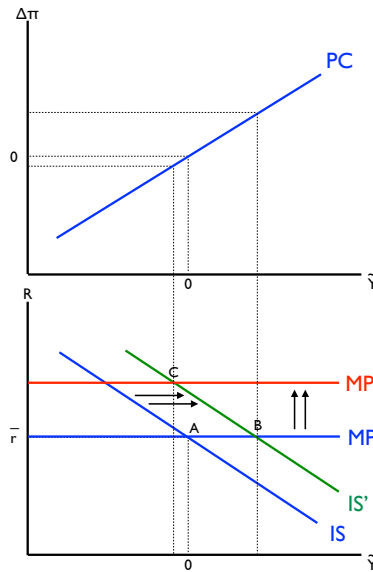
$$R = R^{ff} + \bar{p} \quad (5)$$

$$\Delta\pi = \bar{v}\tilde{Y} + \bar{o} \quad (6)$$

where R is the real interest rate paid by borrowers; $\bar{r}$ is the marginal product of capital; R^{ff} is the real federal funds rate; $\bar{p}$ is the risk premium charged by private lenders; π is the inflation rate; and $\bar{o}$ is an external inflation shock.

Imagine that it is 2002, and the economy has emerged from the recession of 2001. Generation X'ers feel optimistic about the future, and they increase spending on housing as a share of permanent income, racking up debt.

- (a) (6 points) Illustrate this development *and any response by the Federal Reserve* in the IS-MP and Phillips Curve diagrams below. Label and describe what you've drawn. *If a curve shifts twice in this answer, show that.*



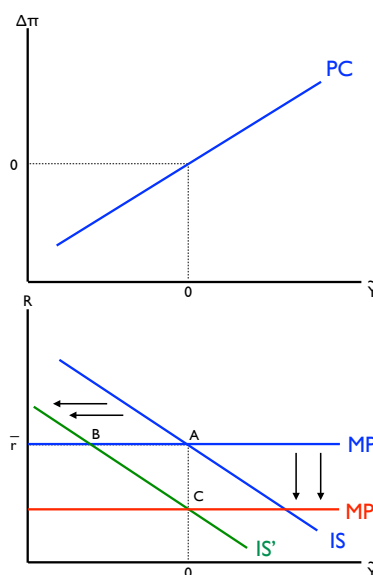
Answer: The optimism of Gen X and their increased spending on housing show up as a rightward shift in the IS Curve. This would shift the economy from point A to point B in the graph, which would increase inflation through the Phillips Curve.

In response, the Fed would act to pull inflation back down, by shifting the MP Curve upward as shown, moving the economy to a place like point C, which as I've drawn it here is actually in negative $\tilde{Y}$ territory. If the Fed needs to bring inflation back down, it might have to raise interest rates that high. If it acts quickly, it might aim for a point C at a $\tilde{Y}$ closer to zero, like at the original point A.

Now imagine that it is 2006, and a major crisis hits the housing market. Generation X'ers discover that their incomes are not sufficient to make their mortgage payments on the enormous houses they starting buying in 2002.

Homeowners sharply reduce their consumption spending in hopes of avoiding default on their mortgages. But it's too little, too late.

- (b) (6 points) Illustrate this development *and any response by the Federal Reserve* in the IS-MP and Phillips Curve diagrams below. Label and describe what you've drawn. *If a curve shifts twice in this answer, show that.* [You don't need to show the events from part (a).]



Answer: If homeowners reduce their consumption spending, that would shift the IS Curve leftward, which moves the economy to a place like point B from point A, if we started from the short-run equilibrium at point A.

This would cause a reduction in inflation, and the Fed would act to increase inflation by shifting the MP Curve downward, moving the economy to a place like point C. I've drawn point C as directly below

point A, and the Fed might instead have to reduce R even more in order to produce a boom and enough inflation to raise π back to the target.

Despite their efforts, many Generation X'ers default on their mortgages, and lenders lose their loaned funds. Unfortunately, those “lenders” aren't just a few banks. Because the original mortgage lenders sold those mortgage agreements off to many different banks and investors, many agents in the economy lose assets. Even worse, remaining lenders don't know exactly which agents lost a lot in the mortgage market, only that many of them did and thus are not good borrowers because they're near bankruptcy.

- (c) (6 points) If you were a good lender during this period, with funds available, and a borrower walked into your office and asked to borrow money, what interest rate would you charge him or her? Would it be higher, lower, or the same as the usual interest rate you'd charge, and why?

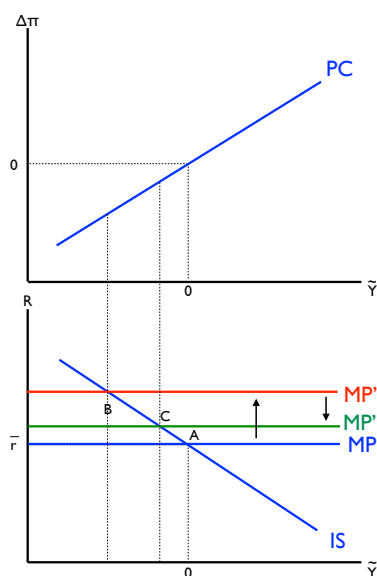
Answer: There's a simple way and a complicated way of looking at this. The simple way is to observe that if the mix of borrowers now includes desperate and impoverished borrowers who are likely to be bad loans, a smart lender will charge a premium in hopes of breaking even given the higher probability that a borrower will default.

The complicated point is that this is a classic case of “adverse selection,” in which borrowers with private information about their own balance sheets are likely to differentially approach lenders. The borrowers most eager to borrow are probably those who are the most desperate! A smart lender might even decide to leave the market altogether rather than face the prospect of lending to somebody who is almost surely a bad borrower.

- (d) (6 points) Illustrate this development *and any response by the Federal Reserve* in the IS-MP and Phillips Curve diagrams below. Label and describe what you've drawn. *If a curve shifts twice in this answer, show that.* [You don't need to show the events from parts (a) or (b).]

Answer: You need to show *two shifts* in the MP Curve here. The first is caused by the reaction of lenders to the problems with borrowers' balance sheets. If lenders charge a higher risk premium on loans, then the MP Curve will shift upward.

This causes a reduction in inflation at a point like B, and the Fed will react by lowering the interest rate and the MP Curve, provided that it still can. I've shown the Fed lowering the MP Curve to a point like C, which isn't all the way back to point A, and ideally the Fed might like to lower R even further if it can.



- (e) (6 points) If the Federal Reserve had already reduced the nominal federal funds rate to zero before the events of part (d), would that change your story at all, and if so how? Explain.

Answer: If the Fed had already reduced the nominal federal funds rate to zero before part (d) happened, then it couldn't lower R any more at all. To be technical about it, the events of part (d) imply a further reduction in inflation and even an *increase* in the real federal funds rate! But what's important here is that if the Fed has reached the zero lower bound on the nominal federal funds rate, it can't lower interest rates any more, and the events of part (d) would cause a recession and disinflation that the Fed couldn't counterattack (without some other policy tool!).

- (f) (6 points) What else could the government use to address the issues afflicting the economy in part (d) other than the Federal Reserve's monetary policy? Illustrate this alternative in the IS-MP and Phillips Curve diagrams below. Label and describe what you've drawn. [You *should* show the events from part (d) but not from parts (a) or (b).]

Answer: The other tool the government has at its disposal is *fiscal policy*. By increasing government spending on goods and services, it could shift the IS Curve to the right and help push the economy back toward a boom and reflation. In the graph, this shows up as point D, at the intersection of a rightward-shifted IS' and a still-elevated MP'' .

